

Product Requirements Document (PRD): Symptom Checker Web Application

Version: 1.0

Status: Draft

Target Audience: Engineering, Design, Stakeholders

1. Executive Summary

The **Symptom Checker Web Application** is a patient-centric digital health tool designed to bridge the gap between experiencing physical discomfort and understanding potential medical conditions. By providing an intuitive, interactive interface for symptom reporting, the platform delivers data-driven health insights, precautionary measures, and educational resources.

***Disclaimer:** This tool is for educational purposes only and is not a substitute for professional medical advice, diagnosis, or treatment.*

2. Project Objectives

- **Accessibility:** Provide a low-barrier entry point for users to screen their health symptoms.
 - **User Experience:** Deliver a “beautiful,” calming, and frictionless interface that reduces user anxiety.
 - **Informed Decision Making:** Empower users with probable conditions and actionable precautionary advice.
 - **Data Integration:** Leverage Kaggle-sourced medical datasets (e.g., *Disease Symptom Description Dataset*) containing 700+ unique diseases and 300+ symptoms to power a robust prediction engine.
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3. Target Audience

User Segment	Primary Need
Health-Conscious Individuals	Quick screening of minor symptoms to decide if a doctor's visit is necessary.
Caregivers	Assessing symptoms for family members (children, elderly) to understand urgency.
Students/Researchers	Exploring the intersection of web development and predictive healthcare analytics.

4. User Experience (UX) & Interface (UI) Strategy

To meet the user's request for a "beautiful interface" and "best user experience," the design will follow these core principles:

4.1 Visual Design Language

- **Color Palette:** A "Healing & Trust" palette.
 - **Primary:** Soft Teal (#2D9CDB) or Serene Blue (#0052CC) for trust and professionalism.
 - **Secondary:** Mint Green (#27AE60) for health and growth.
 - **Background:** Clean White (#FFFFFF) or ultra-light Gray (#F8F9FA) to ensure a clinical, hygienic feel.
- **Typography:** Sans-serif fonts (e.g., Inter, Montserrat) for high readability across devices.
- **Imagery:** Use of soft-edged icons and minimalist illustrations instead of clinical, graphic medical photos.

4.2 UX Flow (The "Frictionless Path")

1. **The "Safety First" Onboarding:** A brief, 3-step introduction that builds trust. Includes a mandatory "I Understand" toggle for the medical disclaimer and a progress bar to show the user's journey.

2. Intelligent Symptom Selection:

- **Visual Selection:** An interactive 3D or 2D “Human Body Map” where users can click on the affected area (e.g., Head, Chest, Abdomen).
- **Smart Search:** A search bar with medical-grade autocomplete that suggests symptoms as the user types (e.g., typing “Fe” suggests “Fever”, “Fear”, “Feeling cold”).

3. Contextual Refinement: Dynamic, one-at-a-time questions (e.g., “Is the fever accompanied by chills?”) using a “Card-Swipe” or “Large Button” interface to minimize cognitive load.

4. Empathetic Results Dashboard:

- **Prioritization:** Top 3 probable conditions displayed with a “Confidence Score” (e.g., 85% match).
 - **Visual Triage:** Color-coded urgency levels: **Red (Seek Emergency Care)**, **Orange (Consult a Doctor)**, and **Green (Home Care/Monitor)**.
 - **Actionable Next Steps:** Specific precautionary measures (e.g., “Stay hydrated,” “Avoid strenuous activity”) and a “Download Report” button for their doctor.
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5. Functional Requirements

5.1 Core Features

ID	Feature	Description
FR-01	Symptom Input	Users can select multiple symptoms (Fever, Cough, Nausea, etc.) via a multi-select search or category cards.
FR-02	Prediction Engine	A rule-based or ML-trained model (scikit-learn) that matches symptoms to the Kaggle dataset.
FR-03	Result Dashboard	Displays top 3 probable diseases with percentage likelihood or confidence levels.
FR-04	Precautionary Guide	Provides actionable advice (e.g., “Stay hydrated,” “Avoid dairy”) based on the predicted condition.
FR-05	Medical Disclaimer	A non-skippable modal or footer-anchored disclaimer regarding the tool’s educational nature.

5.2 Technical Stack

- **Frontend:** React.js or Next.js with Tailwind CSS for a modern, responsive interface.
 - **Animations:** Framer Motion to provide smooth, “beautiful” transitions between symptom selection steps.
 - **Backend:** Python with FastAPI for high-performance API endpoints and automatic Swagger documentation.
 - **Machine Learning:** Scikit-learn (Random Forest or Naive Bayes) for disease prediction based on symptom weights.
 - **Data Handling:** Pandas for cleaning and processing the Kaggle CSV datasets.
 - **Database:** PostgreSQL for persistent storage of user session logs (anonymized) and precautionary content.
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6. Non-Functional Requirements

- **Responsiveness:** The application must be “Mobile-First,” ensuring 100% functionality on smartphones.
 - **Performance:** Prediction results should be delivered in under 2 seconds.
 - **Privacy:** No Personal Health Information (PHI) should be stored without explicit user consent (GDPR/HIPAA considerations).
 - **Accessibility:** Compliance with WCAG 2.1 Level AA standards (color contrast, screen reader support).
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7. Roadmap & Future Enhancements

- **Phase 1 (MVP):** Basic symptom selection and rule-based prediction.
 - **Phase 2:** Integration of a Machine Learning model for higher accuracy.
 - **Phase 3:** “Find a Doctor” integration using geolocation APIs.
 - **Phase 4:** Voice-activated symptom reporting for improved accessibility.
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8. Success Metrics

- **User Retention:** Percentage of users who complete the symptom check flow.
- **Accuracy Feedback:** User-reported accuracy of the suggested conditions.
- **Engagement:** Average time spent on the “Precautionary Advice” page.